

1. Introduction

"Good morning/afternoon.

Our group number is 3, and our project title is *Implementation of OSPF for Scalable Routing in an Enterprise Network*."

2. Objective

"The main objective of our project is to design and configure a network using OSPF routing protocol, and to demonstrate how it provides efficient and scalable routing in an enterprise network."

3. What is OSPF

"OSPF stands for Open Shortest Path First.

It is a link-state routing protocol used to find the best path between networks.

It builds a complete map of the network and uses Dijkstra's algorithm to calculate the shortest path."

4. Difference from Distance Vector

"Unlike distance-vector protocols like RIP, OSPF does not send full routing tables periodically.

Instead, it sends updates only when there is a change in the network.

This makes OSPF faster and more efficient."

5. Network Topology

"In our project, we designed a topology with four routers: R0, R1, R2, and R3.

Each router is connected to a separate LAN, and routers are interconnected using serial links.

We also used a shared network to demonstrate DR and BDR election."

6. OSPF Configuration

"We configured OSPF on all routers using process ID 1.
All networks are placed in Area 0, which is the backbone area."

7. Wildcard Mask

"We used wildcard masks in network configuration.
A wildcard mask is the inverse of a subnet mask.
For example, subnet mask 255.255.255.0 becomes wildcard mask 0.0.0.255."

8. Verification

"To verify our configuration, we used the following commands:
'show ip ospf neighbor' to check neighbor relationships,
'show ip route' to check routing tables,
and 'show ip protocols' to confirm OSPF configuration."

9. DR and BDR

"In multi-access networks, OSPF elects a Designated Router and a Backup Designated Router.
The DR acts as a central point for communication, while the BDR acts as a backup.
This reduces network traffic and improves efficiency."

10. DR Election

"The DR and BDR are elected based on priority and router ID.
The router with the highest priority becomes DR.
If priorities are equal, the router with the highest router ID becomes DR."

11. Area 0

"Area 0 is the backbone area in OSPF.
All other areas must connect to Area 0.
This hierarchical design improves scalability and reduces routing overhead."

12. Key Features of OSPF

"OSPF provides fast convergence, efficient routing using cost metric, reduced traffic using DR and BDR, and scalability through area design."

13. Challenges

"During our project, we faced some challenges such as incorrect IP addressing, interfaces being down, and OSPF neighbors not forming.
We solved these issues by verifying configurations and correcting errors."

14. Conclusion

"In conclusion, our project successfully demonstrated OSPF implementation in a multi-router network.
OSPF provides fast, efficient, and scalable routing, making it suitable for enterprise networks."

15. Closing

"Thank you."
